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Enhancing Network Intrusion Detection with Neural Networks: A Comparative Study of ANN, LSTM, and MLPC Models

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Abstract

As network security threats continue to grow in complexity, the development of robust intrusion detection systems (IDS) has become essential. This study proposes an IDS framework leveraging three types of neural networks: Artificial Neural Network (ANN), Long Short-Term Memory (LSTM), and Multilayer Perceptron Classifier (MLPC) to enhance detection accuracy across various network intrusion scenarios. Our methodology involves comprehensive data preprocessing and a comparative analysis of binary and multiclass classification models. The selection of each neural network is motivated by their respective strengths: ANN for handling complex patterns, LSTM for capturing temporal dependencies, and MLPC for efficient multiclass classification. Our results demonstrate significant improvements in accuracy, with ANN, LSTM, and MLPC models achieving 99.47%, 97.45%, and 99.54% accuracy, respectively. These findings underscore the effectiveness of neural network approaches in improving IDS performance and provide a foundation for further exploration of machine learning applications in cybersecurity.

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Keywords: Network intrusion detection, IDS, Neural Network, ANN, LSTM, MLPC, Machine learning, Cybersecurity

1. Introduction

In today's digital world, network security is more important than ever. As cyber threats and intrusion attempts grow more sophisticated, it is essential to quickly and accurately detect malicious activities within network traffic to protect systems and sensitive data [1, 2].

The Conventional intrusion detection solutions have demonstrated a number of issues, including low detection accuracy, low detection efficiency, and other mediocre detection performance in real networks [3]. Novel approaches that combine network intrusion detection with artificial intelligence have been put forth in the literature as solutions to these issues. Applications of neural networks and neural algorithms to intrusion detection are referred to as artificial intelligence-based network intrusion detection technology. This automation of intrusion detection model detection is made possible by the application of AI techniques

[4, 5, 6]. In [7], The author suggests using neural networks (NNs), more especially backpropagation NNs, to identify intrusions in Tor networks. To transport data over the Tor network, the authors built a simulation system that consists of a Deep Web browser (client) and a Tor server. They acquired data for the Backpropagation NN to estimate using Wireshark Network Analyzer. The system's efficacy for intrusion detection in Tor networks is demonstrated by the simulation results. In [8] To identify and block attacks on the network, the authors in this paper propose an IDS system; Intrusion detection system that is deployed on vehicle terminals. They introduce an IDS based on a convolutional neural network for the Internet of Vehicles, designed to operate efficiently on low-powered embedded vehicle terminals for real-time monitoring of in-car network data. Experimental results demonstrate the system's high-accuracy detection and low-latency performance for identifying attack traffic. others techniques are used in [9], The study develops a model based on an optimized convolutional neural network and long short-term memory network for intrusion detection in computer networks using machine learning techniques. The study utilizes KDDCUP99 intrusion data for training and compares the proposed model with traditional decision tree algorithms, convolutional neural networks, and deep neural networks. The results demonstrate a high prediction accuracy of 97.2% and a low test loss of 0.045, showcasing the suggested algorithm's efficacy in significantly improving the classification accuracy of intrusion detection. in [10] This research presents a novel approach to intrusion detection by employing artificial intelligence methods, specifically an enhanced hybrid model that incorporates both a recurrent neural network (RNN) and a convolutional neural network (CNN). The hyperparameters of the model have been fine-tuned, and the NSL-KDD dataset, which includes four distinct attack categories, is utilized for the simulation. The use of an improved neural network for intrusion detection is presented in [11], the author introduces an Intrusion Detection model, IPSO-BPNN, aiming to enhance accuracy by optimizing the neural network through an improved Particle Swarm Optimization (IPSO) algorithm.

Within this context, the KDD Cup 1999 dataset, well-established in the cybersecurity domain, serves as a comprehensive and realistic resource for assessing intrusion detection systems. The dataset incorporates a wide array of attack scenarios and normal network traffic, rendering it a valuable asset for research in the field of security. [12, 13].

Existing intrusion detection systems (IDS) face several key limitations in accuracy, efficiency, and adaptability to new and evolving threats. Conventional IDS solutions often struggle with low detection accuracy and efficiency, particularly in real-time network environments, making it challenging to detect increasingly sophisticated cyber threats. Additionally, many traditional systems rely heavily on predefined rules or signatures, which limits their ability to detect novel or unknown attack types. As a result, there is a growing need for more adaptive, intelligent approaches that can enhance detection rates while maintaining high performance across various network conditions. To address these challenges, our study proposes a neural network-based IDS framework using Artificial Neural Network (ANN), Long Short-Term Memory (LSTM), and Multilayer Perceptron Classifier (MLPC) models. By leveraging the strengths of each model—ANN for pattern recognition, LSTM for temporal sequence processing, and MLPC for handling complex classification tasks—our approach aims to improve detection accuracy, especially in multiclass scenarios. Using the KDD Cup 1999 dataset for training and evaluation, we explore both binary and multiclass classification techniques to optimize intrusion detection performance, achieving robust results that highlight the benefits of neural network integration in IDS.

This paper centers on the application of advanced machine learning techniques like LSTM, ANN and MLPC neural networks to enhance network intrusion detection. Key areas of focus include data preprocessing, model design, and the resulting outcomes. The results give an accuracy of 97.45%, 99.47%, and 99.54% for LSTM, ANN, and MLPC, respectively. This research is structured into four sections. Section 1 elaborates on the methodology employed and outlines the system. Section 2 is dedicated to the discussion of data processing. In Section 3, we delve into the proposed models. Section 4 presents the performance comparison, and the conclusion is encapsulated in Section 5.

2. METHODOLOGY AND SYSTEM

We delve into the theoretical foundations and implementation details of the key machine learning algorithms employed in our network intrusion detection system. We focus on three primary models: Artificial

Neural Networks (ANN), Long Short-Term Memory (LSTM) networks, and Multilayer Perceptron Classifiers (MLPC). Each of these models brings unique capabilities to the task of intrusion detection, and we explore their intricacies in depth.

2.0.1. Artificial Neural Networks (ANN)

Artificial Neural Networks (ANNs) are inspired by the interconnected neurons of the human brain. They use activation functions like sigmoid and ReLU to introduce non-linearity, enabling the network to learn complex patterns. Training ANNs involves using the backpropagation algorithm and gradient descent to optimize the network's performance [14, 15, 16].

Artificial Neural Networks (ANN) are well-suited for intrusion detection due to their ability to learn complex patterns in network traffic and generalize to new threats. Their supervised learning approach allows them to classify normal and malicious activities based on historical data. ANN models, particularly Multi-Layer Perceptrons (MLP), leverage non-linear activation functions and backpropagation to refine their accuracy. Their adaptability enables real-time detection of evolving cyber threats, including DDoS and malware attacks, with high precision. This makes ANN a powerful tool for enhancing network security by identifying anomalies and preventing unauthorized access [17]

2.0.2. Long Short-Term Memory (LSTM)

A long-term memory network (LSTM) is a variant of a recurrent neural network designed to process sequential data. They are effective at achieving temporal dependency on data patterns, making them a useful choice for network intrusion detection. LSTMs are used with special architectures consisting of memory cells, input, output and oblivion gates, as well as activation functions such as sigmoid and hyperbolic tangent to adjust the distribution. They can store both short- and long-term information, making them suitable for data processing in IT security situations. instrumental in capturing temporal dependencies for intrusion detection [18, 19].

LSTM networks are well-suited for detecting temporal patterns in intrusion data because they can effectively capture sequential dependencies in network traffic. Their memory cell architecture allows them to retain important information from past events while forgetting irrelevant data, which is crucial for identifying anomalies and attack patterns over time. This makes LSTM ideal for handling dynamic and time-based intrusion scenarios [20, 21]. Unlike traditional models, LSTMs can remember relevant past events while filtering out noise, making them ideal for identifying evolving attack patterns such as DDoS, slow brute-force attempts, and port scans. Their ability to analyze traffic over time allows them to detect anomalies that static models might miss, enhancing the accuracy of intrusion detection systems by learning normal network behavior and flagging deviations [22].

2.0.3. Multilayer Perceptron Classifiers (MLPC)

Multi-layer perceptron classifiers (MLPCs) are neural networks with many hidden layers, which allow our system to handle complex classification tasks. Their structure includes input, hidden and output levels, which are important for data processing. MLPC uses activation functions such as sigmoid and ReLU to introduce non-linearity into the model. On the training side, we cover the process in detail, including forward and backward passes, and introduce optimisation techniques such as gradient descent [23].

MLPC (Multilayer Perceptron Classifier) is effective for intrusion detection due to its ability to handle complex classification tasks. Its deep architecture with multiple hidden layers allows it to capture intricate patterns and relationships within the network traffic data. By leveraging activation functions like ReLU and softmax, MLPC can efficiently differentiate between normal traffic and various types of attacks, making it highly suitable for multiclass intrusion detection scenarios [24].

3. RESULT AND DISCUSSION

The KDD Cup 1999 database [25] used in our study on network intrusion detection comprises a diverse set of features describing network traffic.

The labels provide a ground truth for training and evaluating intrusion detection models. In our case, it seems the labels have been grouped into the following categories:

- Normal (type: normal): Represents normal, benign network connections.
- Denial of Service (DoS): Attacks known as denial of service (DoS) are designed to prevent people from accessing a computer or network resource.
- Probe: A probe is an attempt to learn more about a network, sometimes used as a lead-up to a more serious attack.
- Remote to Local (R2L): Unauthorized access attempts from a remote site to a local system are represented by the term "Remote to Local" (R2L).
- This denotes attempts to obtain root (administrative) access by beginning as a regular user and trying to exploit security holes. It is essential to comprehend these categories in order to train the model to correctly classify network connections.

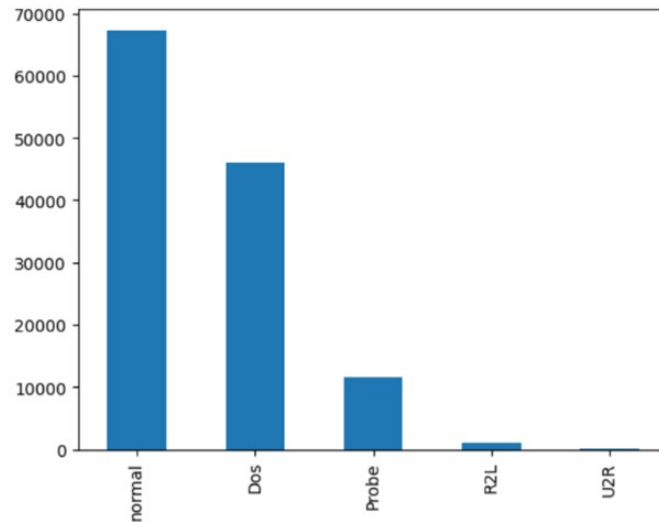


Fig. 1. classification the class of full KDDCUP 99 dataset

3.1. Data Preprocessing

Data preprocessing is a crucial step in our project, aimed at preparing the data for machine learning. The quality and readiness of the data significantly impact the performance of our models. Figure 1 shows the attacks numbers in the full KDD CUP 1999 dataset.

Normalization: Data normalization was applied to scale the features and make them comparable. This step is vital because features may have different scales, which can introduce biases into the model's learning process.

$$x_{Normalized} = \frac{x - means(x)}{std(x)} \quad (1)$$

Where $x_{Normalized}$ is the normalized value of feature, the mean(x) is the mean of the values of feature x in the dataset. and std(x) presents the standard deviation of the values of feature x in the dataset.

Before applying one-hot encoding (Fig.2), the protocol_type feature was represented by categorical labels such as "tcp," "udp," and "icmp". After applying one-hot encoding (Fig.3), the protocol_type feature

has been transformed into multiple binary columns, each representing a distinct category. One-Hot Encoding: Given the presence of categorical features like protocol type and service, we employed one-hot encoding to convert them into numerical representations. This allowed us to retain categorical information while enabling the models to work with the data effectively.

protocol_type	
0	tcp
1	udp
2	tcp
3	tcp
4	tcp

Fig. 2. Before One-Hot Encoding

	protocol_type_icmp	protocol_type_tcp	protocol_type_udp
0	0	1	0
1	0	0	1
2	0	1	0
3	0	1	0
4	0	1	0

Fig. 3. After One-Hot Encoding

The number of features has been increased to 125. The feature count has been expanded to 125, up from the original 41, following the application of one-hot encoding to the dataset. In this paper, 75% of the data is designated for the training process, while the remaining 25% is reserved for testing. rather than 41 features after executing one hot encoder on the dataset.

3.2. Proposed models

Within this section, we provide a comprehensive overview of the machine learning models proposed for network intrusion detection, elaborating on their details and discussing their performance. In this study, accuracy was selected as the primary evaluation metric due to its simplicity and effectiveness in assessing the overall performance of the models on the KDD Cup 1999 dataset. Given that the dataset is relatively balanced and widely used in the field of intrusion detection, accuracy provides a straightforward and interpretable measure to compare the performance of different neural network models. Additionally, the main objective of this work was to conduct a comparative study between ANN, LSTM, and MLPC models, where accuracy serves as a common and standardized metric for fair evaluation.

For binary classification, the accuracy was computed based on the ability to distinguish between normal traffic and attack traffic. The performance metric used is binary cross-entropy loss with the Adam optimizer, and the accuracy was measured by comparing the predicted class with the true class.

For multiclass classification, the accuracy was calculated based on the model's ability to correctly classify traffic into five categories: Normal, DoS, Probe, R2L, and U2R. The categorical cross-entropy loss function was used along with the softmax activation function for the output layer.

The selection of hyperparameters such as 20 epochs and a batch size of 32 was based on empirical experimentation to balance performance and training stability. 20 epochs allowed the models to effectively learn patterns without overfitting, while 32 as a batch size ensured smooth convergence and stable gradient updates. These settings were validated through multiple trials, leading to high accuracy in both binary and multiclass classification tasks.

3.2.1. Binary classification (Normal and attack)

- LSTM

It comprises an input layer, succeeded by an LSTM layer featuring 50 neurons and a sigmoid activation function. The output layer is composed of a single neuron with a sigmoid activation function. During the training phase, the model is compiled using binary cross-entropy loss and the Adam optimizer. Training is conducted on the preprocessed dataset for 20 epochs, employing a batch size of 5000. A validation split of 20% is implemented throughout the training process. In terms of performance, the LSTM model exhibits notable results, boasting a test accuracy of around 97.45%. This underscores its robust capabilities in effectively identifying network intrusions.

The Fig.4 illustrates the model's accuracy over multiple epochs during the training process. The accuracy on the training dataset is illustrated by the blue line, whereas the accuracy on the test dataset is represented by the orange line.

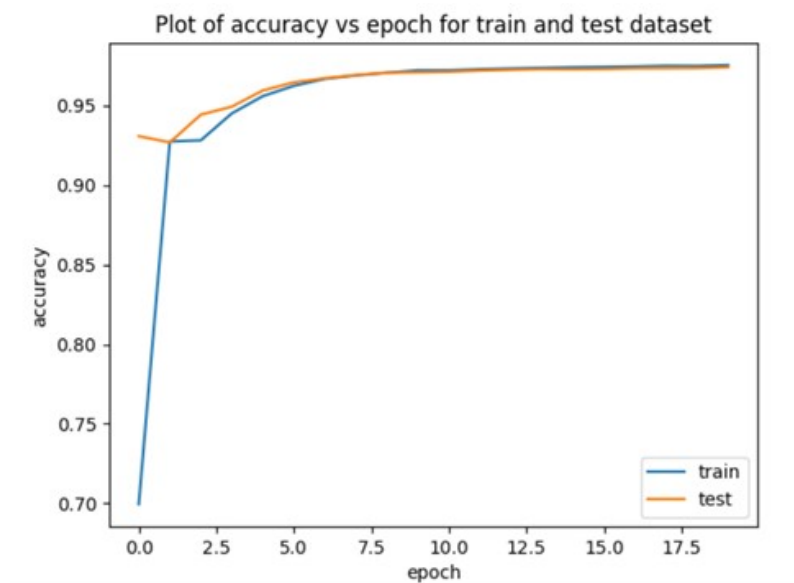


Fig. 4. Training and Test Accuracy Over Epochs

- ANN

It includes multiple dense layers with ReLU activation functions. The final output layer employs the sigmoid activation function, which is well-suited for binary classification. The model is compiled using binary cross-entropy loss and the Adam optimizer. Throughout the training process, it undergoes 20 epochs with a batch size of 32, and a validation split of 20% is employed. The ANN model achieves remarkable results with an impressive test accuracy of approximately 99.55%. It excels in accurately classifying binary network activities and demonstrates robustness.

Figure 5 provides a detailed perspective on the training and validation accuracy across epochs. The blue dots denote the training accuracy, and the solid blue line reflects the validation accuracy. This graph allows a more granular examination of the model's performance during each epoch.

3.2.2. Multi-class classification

- MLPC

It incorporates a dense input layer comprising 50 neurons with a ReLU activation function. The output layer is composed of 5 neurons utilizing a softmax activation function, which is well-suited for multiclass

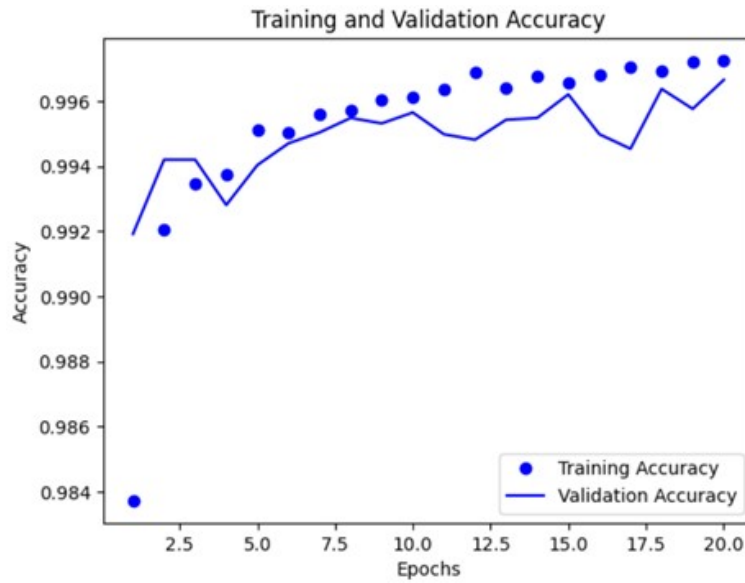


Fig. 5. ANN model's for binary classification

classification. The model is configured with categorical cross-entropy loss and utilizes the Adam optimizer. It undergoes 20 training epochs, employing a batch size of 32, and incorporates a validation split of 20%. The MLPC model achieves impressive results with a test accuracy of approximately 99.54%. It excels in accurately classifying network activities and demonstrates robustness.

Figure 6 illustrates the accuracy of the model over different epochs for both the training and test datasets. The accuracy on the training dataset is depicted by the blue line, while the orange line illustrates the accuracy on the test dataset. This visualization provides an overview of the learning process and convergence of the model.

- ANN

It includes multiple dense layers with ReLU activation functions. For multiclass classification, the final output layer utilizes softmax activation. The model is compiled with categorical cross-entropy loss and the Adam optimizer. It undergoes 20 epochs of training with a batch size of 32 and a validation split of 20%. The ANN model achieves impressive results with a test accuracy of approximately 99.4%. It excels in accurately classifying network activities and demonstrates robustness.

Figure 7 presents a detailed view of the training and validation accuracy throughout the epochs. The training accuracy is represented by the blue dots, and the solid blue line corresponds to the validation accuracy. This graph facilitates a more detailed analysis of the model's performance throughout each epoch.

4. Comparative study

In this section, we provide a synthesis of the proposed model by comparing it with recently published papers. Table 1 encapsulates the performance summary of existing works.

In this work, we conducted a comparative study between ANN, LSTM, and MLPC models for network intrusion detection. Our approach achieved competitive accuracy (99.47% for ANN, 97.45% for LSTM, and 99.54% for MLPC) compared to other works in the literature. This improvement is due to effective data preprocessing, the use of advanced neural network architectures, and hyperparameter optimization.



Fig. 6. Accuracy Over Epochs for Training and Test Datasets (MLPC)

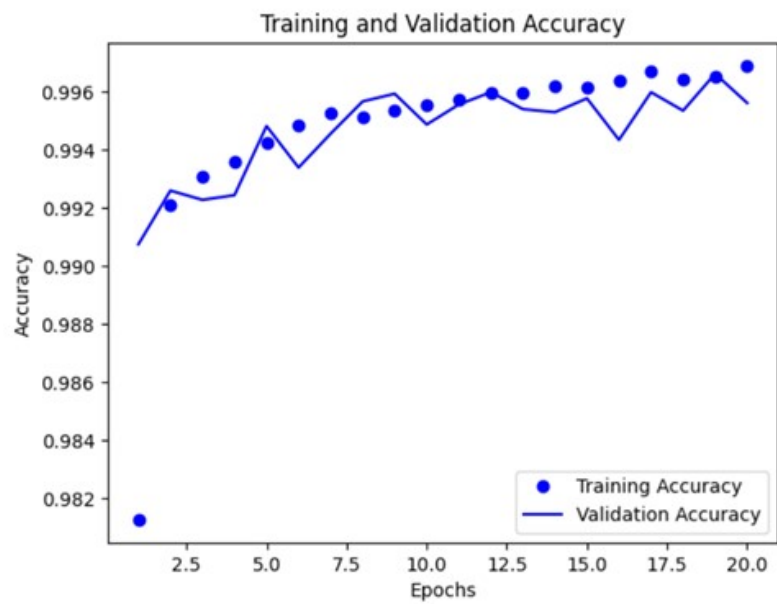


Fig. 7. ANN model for multiclass classification

Table 1. comparative study

Paper	Database	Model	Accuracy (%)
[9]	KDDCUP99	CNN-LSTM	97.2
[10]	NSL-KDD	RNN	-
[26]	KDD CUP 1999	DNN	99.98
[27]	KDD CUP 1999	GA with SVM	97.3
presented work	KDD CUP 1999	LSTM	97.45
		ANN	99.45
		MLPC	99.54

Conclusions

In this study, we proposed a neural network-based Intrusion Detection System (IDS) leveraging Artificial Neural Network (ANN), Long Short-Term Memory (LSTM), and Multilayer Perceptron Classifier (MLPC) models to enhance the detection accuracy of network intrusions. Our approach focused on binary classification (Normal vs. Attack) and multiclass classification to differentiate between different types of attacks (DoS, Probe, R2L, U2R) using the KDD Cup 1999 dataset.

The LSTM model, known for its ability to capture temporal dependencies, demonstrated 97.45% accuracy, making it suitable for detecting sequential patterns in network traffic. The ANN model excelled in recognizing complex patterns, achieving 99.47% accuracy, while the MLPC model showed superior performance in handling multiclass classification with 99.54% accuracy.

In this work, our primary objective is to detect the presence of an intrusion rather than classifying the specific type of attack. Therefore, we focused on binary classification (Normal or Attack) to enhance the detection accuracy of malicious behavior within network traffic. However, we acknowledge the importance of analyzing specific attack types (DoS, Probe, R2L, and U2R), and this will be considered in future work with more details to improve the granularity of the detection system.

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